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AcademicYear: 2025-26	**Important questions of DELD**	**Term–I**
Department: Computer Engineering	**Class : SE Comp**	**Div. : A to F**
Subject: Digital Electronics and Logic Design	**Subject Code: MDM-230-COM**	

Unit I

Boolean Algebra: Basic theorems and properties of Boolean algebra, DeMorgan's rules.

Q1: Simplify following logic function using Boolean algebra

$$f = A\overline{B}C + B\overline{C}D + \overline{A}BC$$

Q2: A staircase light is controlled by two switches, one at the top of the stairs and another at the bottom of stairs.

a) Make a truth table for this system
b) Write the logic equation in SOP form
c) Realise the circuit using AND-OR gates
d) Realise the circuit using NAND gates only

Signed Binary number representation: Signed Magnitude, 1's complement and 2's complement representation.

Q3: Represent +27, - 62 in

a) Signed magnitude
b) 1's complement
c) 2's complement

Q4: Represent +87, - 43 in

a) Signed magnitude
b) 1's complement
c) 2's complement

Simplification Techniques: Sum of product and Product of sum form, Minimization of SOP and POS using K-map.

Q5: Convert following expression to canonical SOP form and minimize it using K-Map

Y = AB + AC + BCD

Q 6: Convert following expression to canonical POS form and minimize it using K-Map

Y = (A+C+D)(B+C+D)(B+D)

Minimization of Boolean function using K-map (up to 4 variables).

Q7. Minimize the following functionin SOP form using K-Map:

$f(A, B, C, D) = \sum m\ (0, 4, 5, 10, 11, 13, 15)$

Q8. Minimize the following function in POS form using K-Map:
$F1(A, B, C, D) = \prod M(0, 1, 2, 3, 11, 12, 14, 15)$
Q9. Minimize the following function in SOP form using K-Map:
$F1(A, B, C, D) = \sum m(0, 4, 5, 10, 11, 13, 15) + d(6, 14)$
Q10. Minimize the following function in POS form using K-Map:
$F1(A, B, C, D) = \prod M(1, 2, 8, 9, 10, 11, 14) \,.\, d(7, 15)$

Unit II

Half Adder, Full Adder:

Q 11: Design Full-adder circuit using two Half-adders. In diagram use block diagram of half-adder.

Half Subtractor, Full Subtractor:

Q 12: Design Full-subtractor circuit using two Half-subtractors. In diagram use block diagram of half-subtractors.

Binary Adder (IC 7483): BCD adder:

Q 13: Design single digit BCD adder using 4-bit binary adder IC7483 and explain two examples of BCD addition.

Look ahead carry generator:

Q 14: Write a short note on look ahead carry generator also explain its importance.

Code Converters: BCD, Excess-3, Gray code, Binary Code:

Q 15: Design BCD toExcess-3code converter using minimum number of gates.

Q 16: Design BCD to Excess-3codeconverterusingminimum Number of NAND gates.

Q 17: Design Excess-3 to BCD code converter using minimum number of gates.

Q 18: Design Excess-3 to BCD code converter using minimum Number of NAND gates.

Q 19: Design 4-bit Binary to Gray code converter using minimum number of gates.

Q 20 : Design 4-bit Gray to Binary code converter using minimum number of gates.

Multiplexers:

Q 21: Implement the expression using 8:1 multiplexer:

$f(A, B, C, D) = \sum m(0, 2, 3, 6, 8, 9, 12, 14)$

Demultiplexers:

Q 22: Implement the following multi-output combinational logic circuit using 4-Line-to-16-Line decoder.

$F1 = \sum m(1, 2, 4, 7, 8, 11, 12)$

$F2 = \sum m(2, 3, 10, 11, 14)$

$F3 = \sum m(11, 12, 14, 15)$

$F4 = \sum m(3, 7, 9, 13)$

Comparators (2 bit):

Q 23: Design 2-bit comparator with minimum number of gates.

Unit III

Flip-Flops: SR, JK, D, T, Preset & Clear, Master Slave JK Flip Flops, Truth Tables and Excitation tables, Conversion from one type to another type of Flop Flop

24. Explain the effect of Preset and Clear signals on the working of Flip Flop.

25. What is a race-around condition in JK FF and how it is overcome in Master Slave (MS)-JK FF?

26. Draw truth tables and Excitation tables of following Flip Flops
i) SR, ii) JK, iii) D, and iv) T

27. Convert the following flip flop
i) JK FF to T FF
ii) JK FF to D FF
iii) SR FF to D FF
iv) SR FF to JK FF

Registers: SISO, SIPO, PISO, PIPO, Shift Registers

28. State different types of Shift Registers and give its applications.

Counters: Ring Counter, BCD Counter, Johnson Counter

29. Write short note on Ring Counter.

30. Write short note on BCD Counter.

31. Write short note on Twisted Ring/Johnson/Moebius Counter.

32. Design MOD-83 counter using IC 7490.

UNIT IV

Algorithmic State Machines: Finite State Machines (FSM) and ASM, ASM charts, notations,

33. Differentiate between FSM and ASM

34. Draw and explain various notations used in ASM charts.

Construction of ASM chart and realization for sequential circuits.

35. Draw the ASM chart for a 2-bit UP-DOWN counter with mode select line M such that: M = 0 (UP Counting), M = 1 (DOWN Counting)

36. Draw the ASM chart for a 2 bit Binary counter having one enable line Esuch that: E = 1 (counting enabled), E = 0 (Counting disabled)

Logic Families: Classification of logic families: Unipolar and Bipolar Logic Families,;

37. Classify logic families used in digital circuits

Characteristics of Digital ICs: Fan-in, Fan-out, Current and voltage parameters, Noise immunity, Propagation Delay, Power Dissipation, Figure of Merits, Operating Temperature Range, power supply requirements.

38. Define the following terms and mention its standard values for TTL family:
i. Fan-in,
ii. Fan-out,
iii. Current parameters
iv. Voltage parameters,
v. Noise immunity/Margin
vi. Propagation Delay,
vii. Power Dissipation,
viii. Figure of Merits,
ix. Operating Temperature Range,
x. Power supply requirements.

Introduction to TTL & CMOS, Comparison between TTL and CMOS

39. Differentiate between TTL and CMOS logic families

Unit V

40.What is the difference between PAL and PLA.

41.Implement following Boolean function using PAL

Fl = Σm (0,2,4,6,8,12)
F2 = Σm (2,3,8,9,12,13)
F3 = Σm (l,3,4,6,9,11,12,14,15)

42.Draw a block diagram of the PLA device and explain.

43.Draw and explain the general structure of PLA.

44.Implement following Boolean function using PAL

F(A, B, C, D) = Σm(0, 1, 3, 15)

45.Implement BCD to Ex-3 code converter using PAL.

46.Implement the following Boolem function using PAL.

F1= Σ m(0,3,5,7,9,10,11,14,15)
F2= Σ m(2,3,12,14)

47.lmplement following Boolean functions using PAL:

FI(A, B, C, D) = Σ m (0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)
F2(A, B, C, D) = Σ m (1,2, 8, 12, 13)

48.Implement gray to binary code using PLA.

49.Implement the following Boolean functions using PAL

X1(A, B, C, D)= Σ m (0,2,6,7,8,9,12,13)

X2(A, B, C, D)= Σ m (3,6,7,11,14,15)

50.Implement the following using PROM

A(X,Y,Z)=∑m(5,6,7)

B(X,Y,Z)=∑m(3,5,6,7)

51.Implementing a Combinational Circuit Using a PAL

W(A, B, C, D) = ∑ m(2, 12, 13)
X(A, B, C, D) = ∑ m(7, 8, 9, 10, 11, 12, 13, 14, 15)

$Y(A, B, C, D) = \sum m(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$
$Z(A, B, C, D) = \sum m(1, 2, 8, 12, 13)$

52.Design of Boolean functions using PLA

$F0 = A + B' C'$
$F1 = A C' + A B$
$F2 = B' C' + A B$
$F3 = B' C + A$